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Human Brain Mapping
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Take-Home Assignment-II

I. Parcellation & Matrix Construction

A. Data and Preprocessing: A preprocessed resting-state fMRI dataset from the Human Connectome Project (HCP) was utilized for this analysis. The data were provided as a CIFTI dense time-series file (rfMRI_REST1_LR_Atlas_hp2000_clean.dtseries.nii) which contained 1,200 timepoints (TR = 0.72 s) across 91,282 grayordinates, comprising 29,696 left-hemisphere cortical vertices, 29,716 right-hemisphere cortical vertices, as well as 31,870 subcortical voxels.

B. Parcellation: The Schaefer 400-parcel atlas (i.e., 17-network variant; Schaefer et al., 2018) was further applied to parcellate the cortical surface into roughly 400 regions of interest (ROIs). Each parcel was assigned to just one of the seven canonical Yeo networks i.e, for Visual there were 47 parcels, for Somatomotor there were 70 parcels, for Dorsal Attention there were 52 parcels, for Salience or Ventral Attention there were 67 parcels, for Limbic there were 24 parcels, for Control there were 61 parcels, and for the Default Mode there were estimated 79 parcels. The Mean BOLD time-series were also extracted for each ROI by averaging across all grayordinates within each parcel which further yielded in a 1,200 × 400 time-series matrix.

C. Time-Series Visualization: Figure-1 below, displays z-scored BOLD time-series for 20 representative ROIs i.e., 4 from each of 5 networks (Default, Salience/Ventral Attention, Dorsal Attention, Visual, and Somatomotor). It should be noted that the within-network ROIs tend to exhibit highly correlated low-frequency fluctuations, with major peaks as well as troughs simultaneously co-occurring in time. This within-network synchrony is particularly riveting in the Default Mode and Visual networks. The between-network comparisons unveil a substantially low temporal coherence which is further consistent with the functional specialization of large-scale brain networks, when at rest.

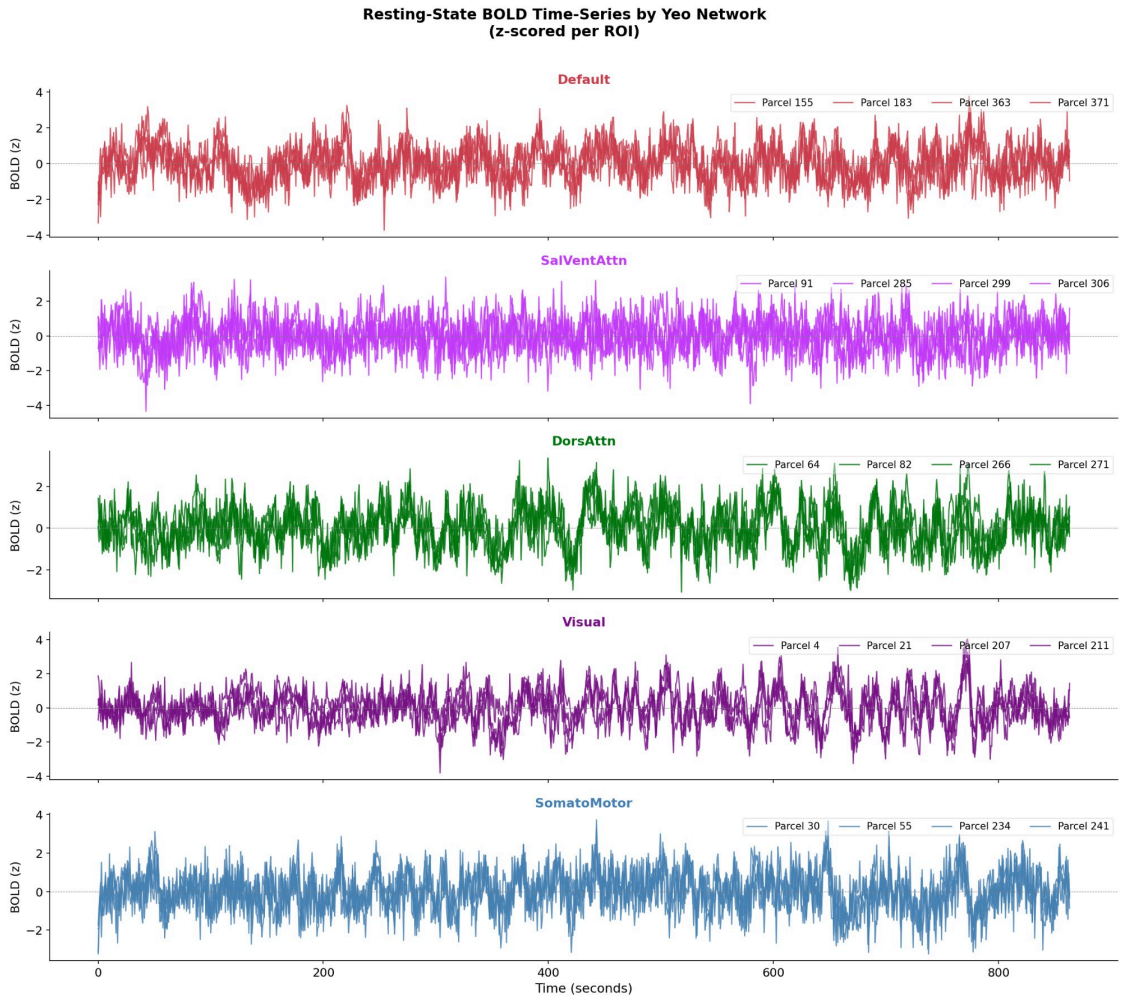


Figure-1. Resting-state BOLD time-series (i.e., z-scored) for 20 ROIs across five Yeo networks. Each subplot here shows 4 parcels from just one network and are color-coded by network identity. Here, the within-network ROIs indicate highly correlated temporal dynamics.

D. Connectivity Matrix Construction: A 400 × 400 functional connectivity matrix was also computed by calculating the Pearson correlation coefficient amidst every pair of ROI time-series. The resulting r-values were later subjected to Fisher's r-to-z transformation in order to further normalize the distribution. Also, notably the diagonal values were set to zero and the z-transformed matrix had a range of −0.484 to 1.376, with a mean off-diagonal value of 0.186.

E. Network Structure Visualization: Figure-2 below presents the connectivity matrix ordered by Yeo 7-network, unveiling a clear block-diagonal structure. The Visual network block here exhibits the strongest within-network correlations (mean z = 0.535, SD = 0.261), further reflecting a highly synchronized activity of primary as well as extrastriate visual cortex during rest. The Dorsal Attention network here also indicates a strong internal coherence (i.e., mean z = 0.412). Notably, the Default Mode Network or DMN, despite being the largest (with 79 parcels),

still shows only a moderate within-network connectivity (i.e., mean z = 0.226) which is only consistent with its distributed spatial extent. The Between-network interactions here are generally weaker, but also notable positive coupling is visible amidst Visual as well as the Somatomotor networks, and also between Control and Default networks. However, here the Limbic network indicates a relatively weak connectivity present both within and between networks, most likely reflecting a lower signal-to-noise in orbitofrontal and temporal pole regions.

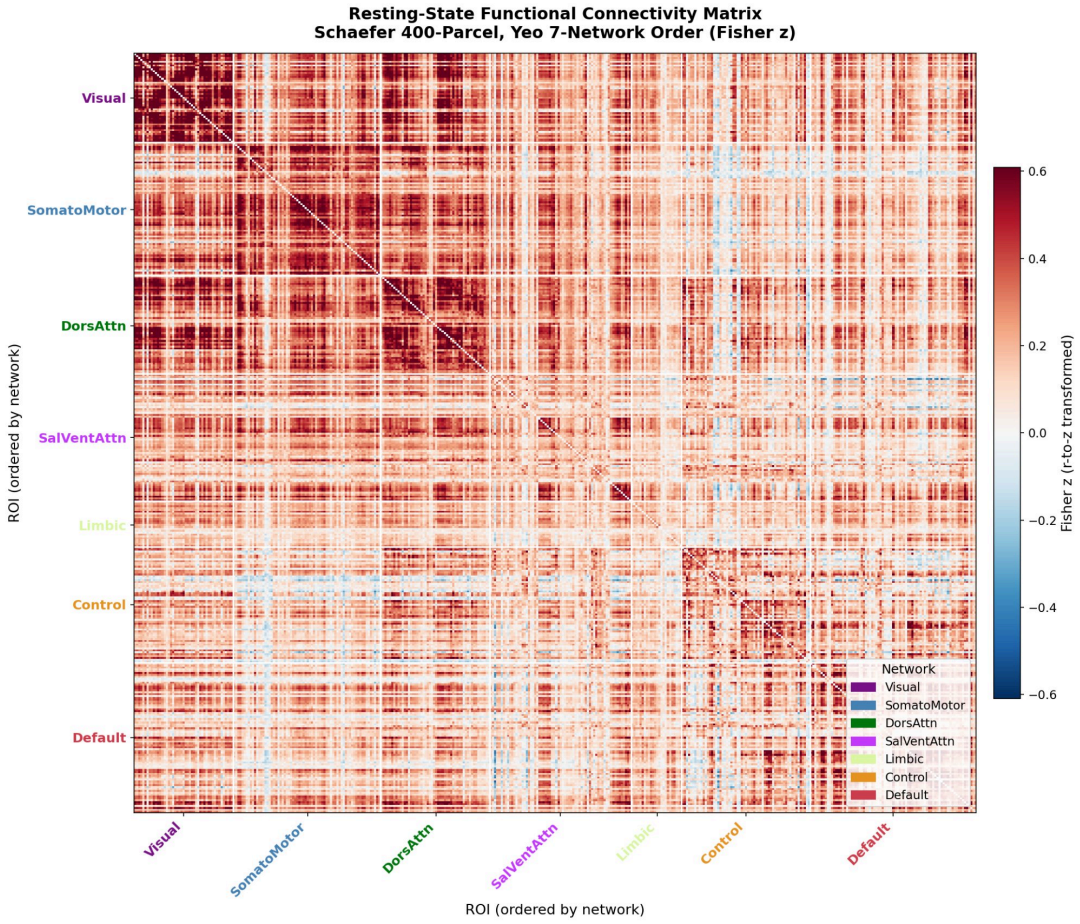


Figure-2. 400 × 400 resting-state functional connectivity matrix (with Fisher z-transformed), ordered by Yeo 7-network. The dashed lines indicate network boundaries whereas the block-diagonal structure reflects within-network coherence across Visual, Somatomotor, Dorsal Attention, Salience/Ventral Attention, Limbic, Control, and Default Mode networks.

II. Lifespan Connectivity Analysis

A. Data Loading and Organization: Pre-calculated full covariance functional connectivity matrices were obtained from the AABC Release 2 dataset (via balsa.wustl.edu). The data used

the Glasser HCP-MMP1 atlas in particular, that comprised of 379 ROIs (i.e., 360 cortical parcels plus 19 subcortical structures). Also notably, each CSV file here encoded one source ROI's covariance with all 379 target ROIs across all subjects. Therefore, for subjects with multiple visits (i.e., V1, V2, V3), only the first visit (i.e., V1) was retained, further yielding a final sample of 1,390 subjects and a 1,390 × 379 × 379 covariance array.

B. Age Group Creation: Subject ages were also obtained from the AABC demographics file (i.e., in particular the age_open column). Later, the four age groups were defined i.e., a) 36–50 (N = 415), b) 50–65 (N = 412), c) 65–80 (N = 334), and d) 80+ (N = 229). The uneven group sizes here reflect the natural difficulty of recruiting older participants, in particular, those aged 80 and above, which should be noted when interpreting the statistical power across groups.

C. Grand-Average Connectivity Matrices: Figure 3 below exhibits the mean connectivity matrix for each age group. Here, the several age-related trends are visible and all groups display the modular block-diagonal structure, which is characteristic of resting-state connectivity, with warm diagonal blocks that further indicate a strong within-network covariance (i.e., sensorimotor, visual, default mode, frontoparietal systems) and cool off-diagonal patches exhibiting an anti-correlated network pairs. Notably, a progressive decline in the overall covariance magnitude is quite evident across the lifespan i.e., the 36–50 group shows the strongest connectivity (with the mean covariance = 141.40 a.u.), declining to 104.42 in the 50–65 group, 73.13 in the 65–80 group, and 54.06 in the 80+ group. This pattern here is further consistent with age-related declines present in functional integration and neural synchrony.

Figure-3. Grand-average functional connectivity matrices (with full covariance) for four age groups, were computed from the Glasser HCP-MMP1 379-ROI parcellation. Here, the yellow dashed lines show the cortical/subcortical boundary whist a progressive decline in covariance magnitude is observed across the lifespan.

D. Contrast Analysis via One-Way ANOVA: A one-way ANOVA was conducted at each edge across the four age groups. Hence, the Bonferroni correction was applied across all 143,641 unique edges ($p < 3.48 \times 10^{-7}$). A total of 77,931 edges (54.25%) survived this stringent correction, that further shows widespread age-related distinctions present in functional connectivity.

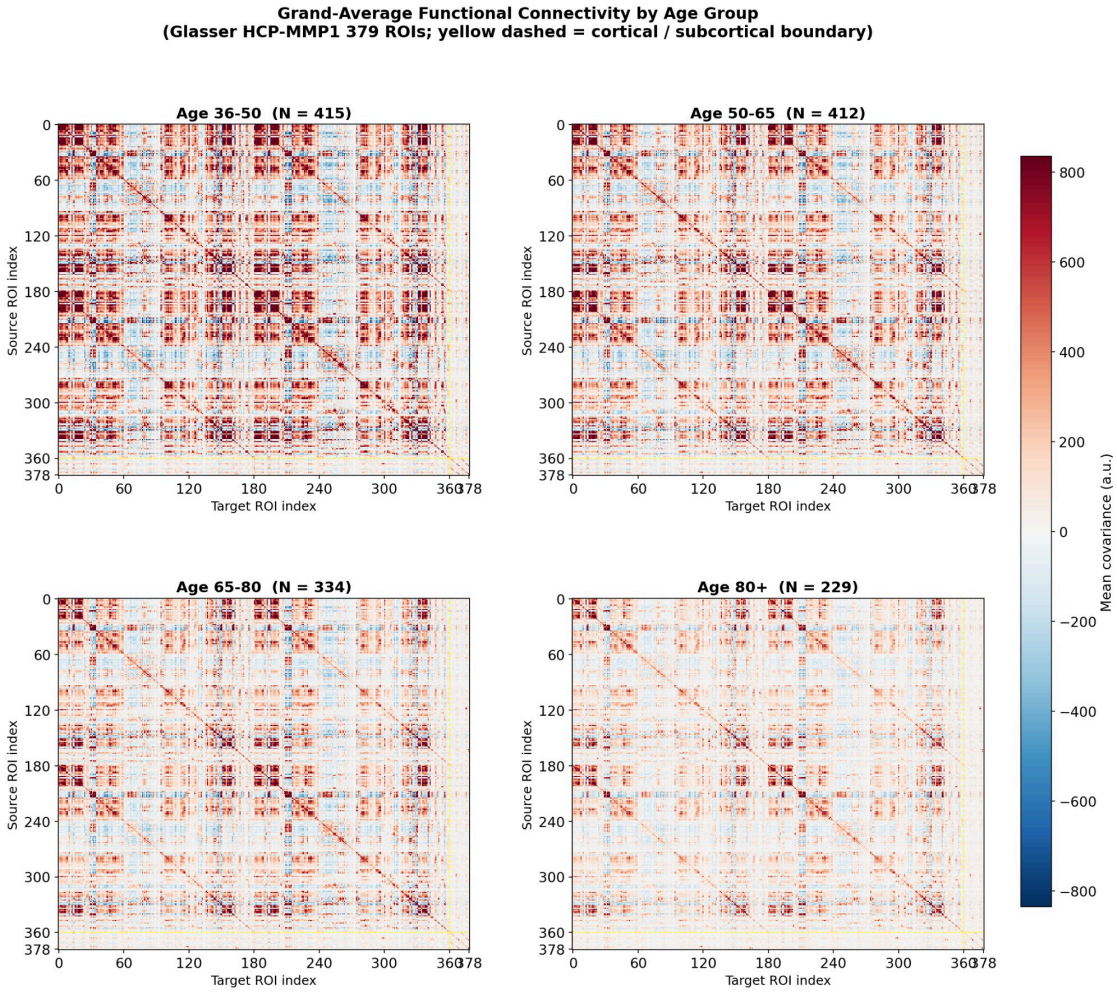


Figure 4 below exhibits the F-statistic matrix wherein the largest F-values appear in connections involving frontoparietal control regions, posterior default mode network (i.e., precuneus, posterior cingulate cortex), as well as the visual-to-higher-association cortex edges. These findings are further consistent with the well-documented age-related regression in top-down executive connectivity as well as the preferential vulnerability of association cortex relative to primary sensory regions. Here, notably the subcortical connections (i.e., ROIs 360–378) also exhibit significant age effects, particularly the thalamocortical and striatal-frontal edges, that further show the contribution of the very subcortical structures to age-related functional reorganization.

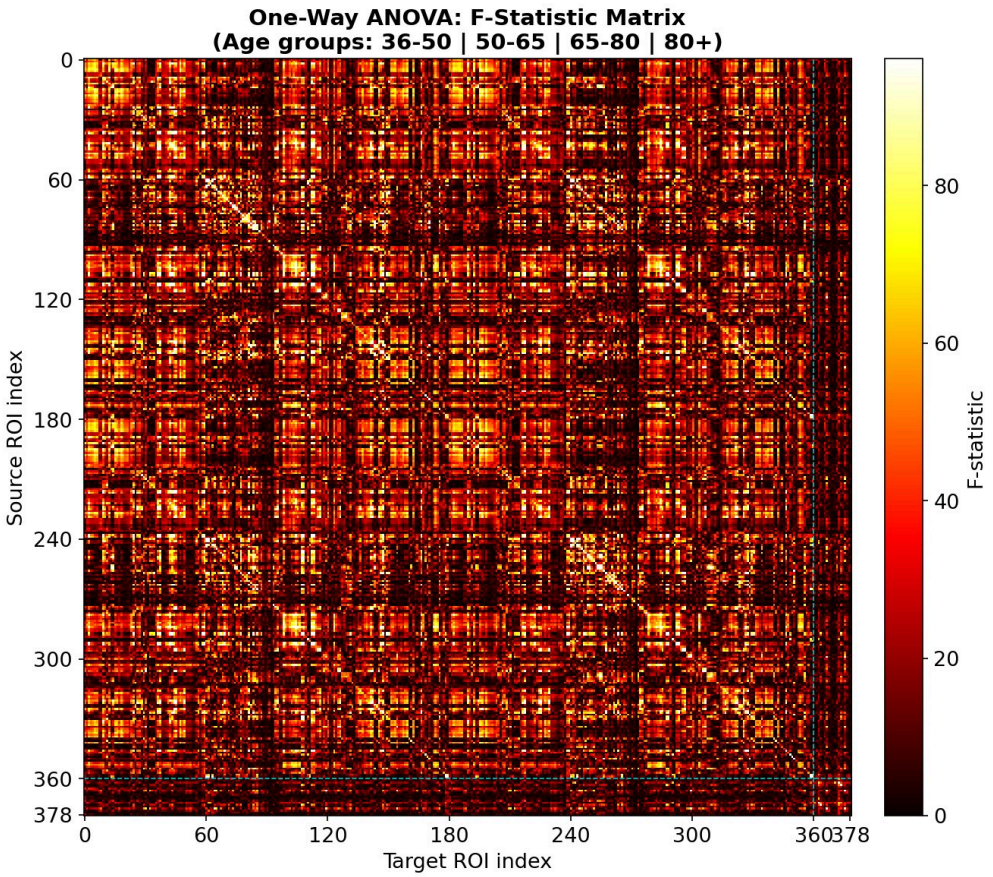


Figure-4. One-way ANOVA F-statistic matrix exhibiting the magnitude of age-related differences present at each edge across four age groups (i.e., 36–50, 50–65, 65–80, 80+). Here, the brighter values show the stronger effects whilst the teal dashed-lines show the cortical/subcortical boundary.

Figure 5 below exhibits the −log10(p) matrix i.e., the green dashed-line on the colorbar point to the Bonferroni-corrected significance threshold (−log10(p) > 6.46). Hence, the values above this line only exhibit edges with statistically significant age differences after multiple comparison correction. Therefore, the widespread presence of bright pixels across the matrix here just confirms that aging profoundly reshapes the very functional connectivity architecture.

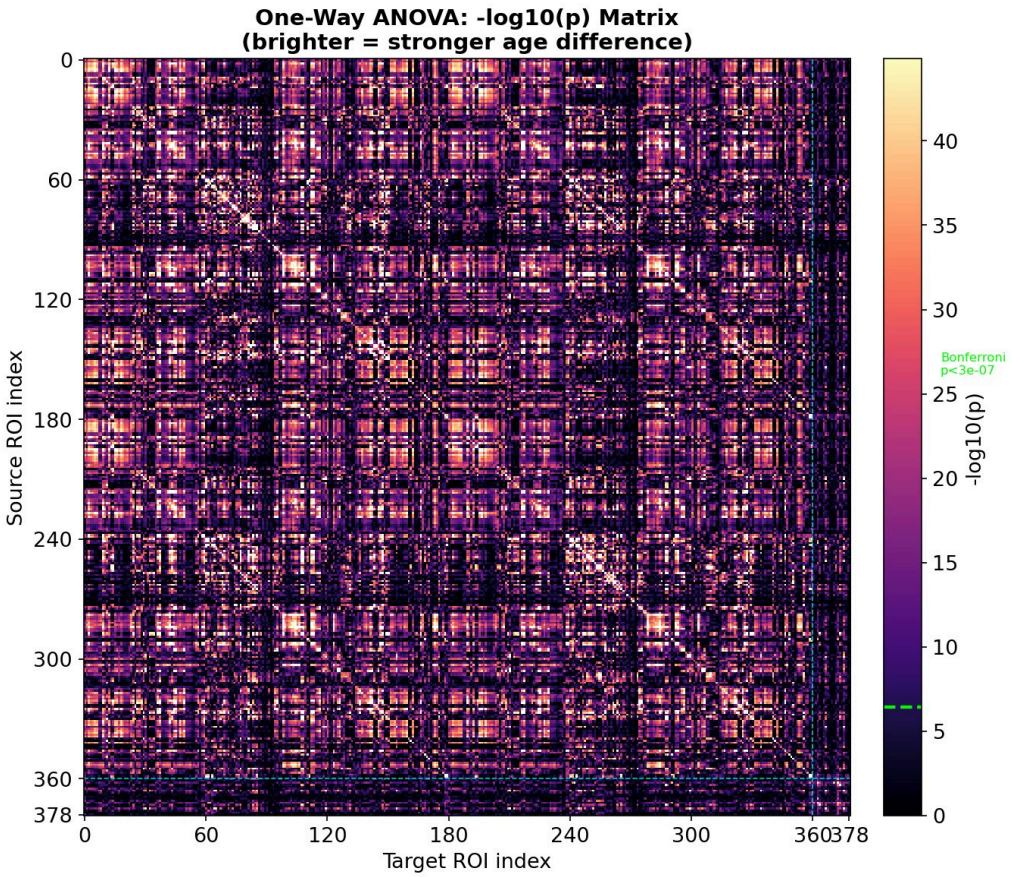


Figure 5. −log10(p) matrix from one-way ANOVA across age groups wherein the brighter pixels indicate stronger statistical significance whilst the green dashed line on the colorbar denotes the Bonferroni-corrected threshold ($p < 3.48 \times 10^{-7}$). Notably, only over 54% of edges survive this correction.

III. Connectivity-Based Predictive Modeling (CPM)

A. Overview and Behavioral Variable: The Connectome-Based Predictive Modeling (CPM) framework (Shen et al., 2017) was further applied to determine whether or not the resting-state functional connectivity could perhaps predict the individual differences present in executive functioning. Therefore, building on findings from Assignment-1, wherein the Flanker task performance exhibited a significant correlation with cortical thickness in the aging population ($r = .14$, $p < .001$), the Flanker executive function raw score (tlbx_fica_rawscore) here was selected as the target behavioral variable. Hence, a total of 1,379 subjects had both complete connectivity data as well as non-missing Flanker scores (range: 12.0–20.0, M = 19.91, SD = 0.64).

B. Feature Selection and Edge Extraction: The upper triangle of each subject's 379 × 379 connectivity matrix was extracted, further yielding 71,631 unique edges per subject. Notably, all edge selection was performed inside the cross-validation loop in order to prevent any kind of circular analysis and/or any inflated accuracy estimates.

C. Leave-One-Out Cross-Validation: For each of the 1,379 LOOCV folds, the following procedure was executed on the N−1 training subjects: a) the Pearson correlation was computed between each edge and Flanker scores; b) the edges with $p < 0.01$ were retained as well as separated into positive ($r > 0$) and negative ($r < 0$) sets; c) the positive and negative network strengths were also computed by summing the respective edge values for each subject; d) a linear regression model was fit here in predicting the Flanker scores from network strength; and e) the held-out subject's Flanker score was also predicted. Hence, across folds, approximately 1,560 positive edges and 1,808 negative edges were consistently selected further exhibiting a stable feature selection.

D. Results: The positive network model achieved a statistically significant out-of-sample prediction ($r = +0.080$, $p = 0.003$), showing that the sum of positively correlated edges can further predict individual variation present in the Flanker performance, albeit modestly. Notably, the negative network model did not reach significance ($r = +0.019$, $p = 0.476$), nor did the combined model which utilized the distinctions of positive and negative strengths ($r = +0.040$, $p = 0.138$). Figure-6 below presents the scatter plots of predicted versus observed Flanker scores for all the three models.

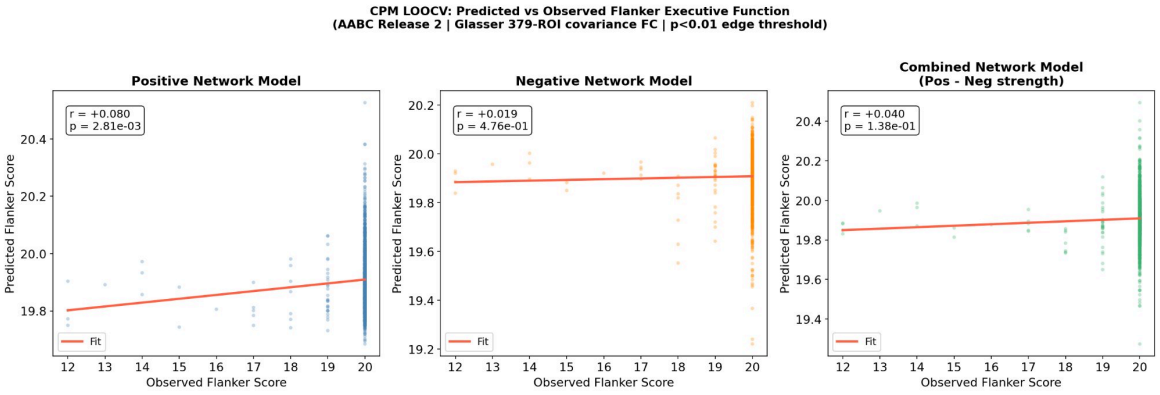


Figure-6. CPM LOOCV results: predicted vs. observed Flanker executive function scores for the positive network model (left), negative network model (center), and combined model (right). The

positive model here achieves a significant prediction (i.e., r = +0.080, p = 0.003). Also, the regression lines and statistics are annotated.

E. Discussion: Here the modest yet significant predictive power of the positive network model is consistent with prior CPM literature that exhibits resting-state connectivity that can capture individual differences present in cognition, even though the effect sizes for executive function in aging populations generally tend to be small. The ceiling effect in Flanker scores (i.e., M = 19.91, with most subjects near the maximum of 20) likely constrains the predictive range. The covariance-based connectivity matrices as opposed to correlation-based, may also contribute to the modest effects, since the covariance retains variance-amplitude information that adds noise when subjects tend to differ systematically in signal scaling. Future work in this realm could further focus on exploring the correlation-based matrices, alternative behavioral targets with greater variability (e.g., Fluid Intelligence), or even partial least squares approaches in order to improve prediction.

References

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